Salus: Web3 Protocol for Incentivized Health Data Sharing and Insurance Discounts

Abstract

Salus is a decentralized Web3 protocol that incentivizes users to share anonymized health data in exchange for insurance discounts and financial rewards. Leveraging blockchain technology, smart contracts, and privacy preserving mechanisms, Salus empowers individuals to take control of their health data while benefiting from token rewards and reduced insurance premiums. Salus employs Trusted Execution Environments (TEEs), decentralized governance, and tokenized incentives to redefine health data sharing in a secure, transparent, and trustless ecosystem.

1. Introduction

The health insurance industry is riddled with inefficiencies, from static risk assessments to privacy concerns. Insurers typically rely on one-time health assessments that fail to reflect an individual's evolving health status. This results in inaccurate pricing models and misses opportunities for users to lower premiums by improving their health. Furthermore, users' health data is siloed within centralized systems, limiting both their control and the potential for data driven innovation.

Key Challenges

Privacy Concerns: Fear of misuse prevents individuals from sharing personal health data.

Outdated Risk Models: Insurers rely on static, one-time health assessments instead of real-time health data.

Lack of User Control: Individuals cannot leverage their own health data to reduce premiums or earn rewards.

Data Siloing: Health data is isolated within centralized databases, stifling innovation in healthcare.

Salus solves these challenges by creating a decentralized, trustless, and user centric platform for health data sharing. With privacy preserving technologies, tokenized incentives, and dynamic health scoring, Salus empowers users to share their health data securely and benefit from lower insurance premiums and cash rewards.

2. The Salus Solution

Salus brings a novel approach to health insurance by combining Web3 technologies, smart contracts, and incentive driven health data sharing into one integrated platform. The goal is to empower individuals to control and monetize their health data, benefit from personalized insurance products, and advance healthcare research.

2.1 Dynamic Health Scoring

Salus collects health metrics from users' devices and health apps like Apple Health, Fitbit, and other IoT devices. The collected data is then processed to generate an onchain health score. This score is dynamic, meaning it reflects real time updates and directly influences users' insurance premiums. The healthier the user, the lower their premium, creating an incentive for individuals to adopt healthier behaviors.

2.2 Privacy Preserving Data Sharing

Salus integrates Trusted Execution Environments (TEEs) through the LIT Protocol. TEEs allow secure and private computation of sensitive data, such as health metrics, without exposing raw data. This ensures that users' personal health data remains private and under their control at all times. Users can decide how much of their data they want to share and with whom, all while benefiting from insurance premium reductions and token rewards.

2.3 Tokenized Incentives

Users are rewarded for contributing anonymized data to a research pool. These contributions power medical research, and in return, users earn Salus tokens or USDC as incentives. High quality and consistent data contributions result in greater rewards, motivating users to stay engaged.

Salus Tokens: Serve as the primary utility and governance tokens within the Salus ecosystem.

USDC Rewards: Users may choose to receive stable coins for immediate value, further incentivizing data sharing.

2.4 Smart Contract Based Policies

Insurance policies within Salus are governed by smart contracts, removing intermediaries and ensuring transparency in insurance processes. Premium adjustments, claims, and payouts are automated and executed through these contracts, reducing administrative costs and streamlining operations.

2.5 Blind Insurance Policies (BIPs)

Salus envisions a future where insurance policies are issued based solely on verifiable, on chain health metrics. This approach, known as Blind Insurance Policies (BIPs), eliminates the need for KYC (Know Your Customer) checks and personal identity verification, thus removing biases such as age, gender, or location. Users are assessed purely based on their health data, making the insurance model more equitable and unbiased.

3. Salus Token Utility

The Salus token is central to the Salus ecosystem, facilitating rewards, governance, and staking.

3.1 Staking for Benefits

Users can stake Salus tokens to access insurance services or to increase their insurance coverage. Staking also allows users to receive additional rewards, such as higher data sharing payouts or enhanced insurance premium reductions.

3.2 Governance through the Salus DAO

Salus token holders can participate in the Salus Decentralized Autonomous Organization (DAO). Token holders can propose and vote on key protocol decisions, including updates to the health scoring algorithms, changes to the reward structure, and future partnerships with insurers and research institutions.

3.3 Earning Tokens

Salus tokens are distributed to users who share anonymized health data with insurers, researchers, and other stakeholders. The token incentives consistent and high-quality data sharing, allowing users to financially benefit from their own health improvements.

4. Architecture and Technology Stack

Salus is built using a modern, scalable tech stack that prioritizes security, privacy, and efficiency.

4.1 LIT Protocol for Data Encryption and Privacy

Salus employs the LIT Protocol to ensure that sensitive health data is securely encrypted and processed in Trusted Execution Environments (TEEs). LIT's decentralized encryption and decryption mechanisms allow Salus to compute health scores while preserving user privacy.

Data Security: The use of TEE ensures that sensitive health data is processed in a secure environment, protecting it from unauthorized access.

Privacy Preserving Computations: LIT enables the generation of health scores without exposing raw health data.

4.2 Privy.io for Authentication and Wallet Integration

Privy.io handles user authentication and wallet integration, ensuring secure and privacy preserving interaction with the Salus platform. Users can connect their crypto wallets to receive token rewards and cashbacks for data sharing.

Secure Authentication: Privy.io enables users to authenticate securely while keeping their data encrypted and off chain.

Wallet Connectivity: Users can interact with blockchain features like rewards and staking without compromising privacy.

4.3 Backend Development with Node.js, Express, and TypeScript

The Salus backend is built using Node.js and Express, with TypeScript for static typing and improved code reliability. This ensures a robust and scalable backend that can handle the complex logic of data processing, health scoring, and smart contract execution.

4.4 Frontend Development with React and Tailwind CSS

The frontend of Salus is developed using React and Tailwind CSS for dynamic content rendering and responsive design. Users can view real time updates on their health scores, insurance premiums, and rewards on a clean and intuitive interface.

4.5 Hosting and Deployment on Vercel

Salus is hosted on Vercel, a platform optimized for seamless deployments and scalability. Vercel ensures that the application remains fast, reliable, and capable of handling growing user traffic.

5. Roadmap

The future of Salus is grounded in its commitment to privacy, user empowerment, and a decentralized approach to health data sharing. However, as the Web3 landscape evolves and adoption of decentralized systems increases, the potential for Salus extends far beyond its initial use case of incentivized health data sharing. This section outlines the ambitious and imaginative roadmap for Salus over the next several years, as well as the broader potential it holds for transforming not just health insurance, but the healthcare and data privacy landscapes as a whole.

5.1. Expansion of Health Device Integration and Data Sources

One of the initial growth strategies for Salus is to integrate more health devices and platforms. Currently, Salus pulls data from widely used health apps like Apple Health and Fitbit, but the next step will be to incorporate a broader range of health tracking devices, wearables, and even medical grade equipment. This expansion will increase the diversity and richness of the data available to insurers and researchers, making the health scores more comprehensive.

Future Vision: Integration with the Internet of Medical Things (IoMT)

The rise of IoMT devices – such as smart glucose monitors, blood pressure cuffs, and remote ECG devices – will enable continuous and granular data collection directly from patients. Salus can facilitate the integration of these devices, providing users with the ability to seamlessly share data from hospital grade equipment to enhance their health scores and insurance offers. The platform could evolve into a global repository for health data, allowing cross border sharing for users who travel or live in multiple countries.

5.2. Partnerships with Insurance Providers, Health Providers, and Research Institutions

Salus aims to establish partnerships with a broader range of stakeholders in the health and insurance ecosystem. Beyond the initial set of insurers, Salus can partner with healthcare providers, pharmaceutical companies, and public health organizations. These partnerships would allow Salus to become a platform for health research on a larger scale, making anonymized, aggregated health data available for research in areas like disease prevention, personalized medicine, and epidemiology.

Future Vision: Salus as a Global Health Data Marketplace

Imagine Salus evolving into a global health data marketplace, where users from around the world can share their anonymized data with researchers, pharmaceutical companies, and insurance firms. Salus tokens or stablecoins could be earned for each approved data request, and users would have full transparency and control over how their data is being used. This marketplace could facilitate breakthroughs in drug development, personalized treatment plans, and even public health initiatives by enabling researchers to access a wealth of high-quality, real-time data.

5.3. Enhanced Privacy and Zero Knowledge Proofs for Data Sharing

As privacy concerns continue to be central to any system that handles sensitive data, Salus will implement more advanced privacy preserving technologies. In addition to using Trusted Execution Environments (TEEs) and LIT Protocol for data encryption, Salus can explore Zero Knowledge Proofs (ZKPs) to ensure that users can prove certain health metrics (like a health score) without revealing the underlying data.

Future Vision: Advanced Privacy Mechanisms

In the future, Salus could employ ZKPbased health scores, where insurance providers can validate health status without ever seeing specific data points like heart rate or blood sugar levels. This would provide users with an unprecedented level of privacy and security while still allowing them to benefit from premium discounts and financial rewards. Multiparty computation (MPC) could also enable secure computations across multiple stakeholders (e.g., insurers, researchers) without any party gaining access to raw user data.

5.4. Introduction of Decentralized Healthcare Services

One long term goal for Salus is to expand beyond insurance and into the provision of decentralized healthcare services. By partnering with telemedicine providers and decentralized healthcare platforms, Salus could allow users to access affordable healthcare services directly through the platform, with payments facilitated via Salus tokens or stablecoins. This integration would close the loop between health data sharing, insurance benefits, and access to healthcare services.

Future Vision: Decentralized Health Networks

Salus could evolve into a broader decentralized health ecosystem, where users not only share data but also access healthcare services, pay for treatments, and even receive diagnostics from decentralized providers. Users could engage with blockchain based health consultancies, access decentralized diagnostic labs, and even receive virtual care through partnerships with global telemedicine platforms. With blockchain enabled identity verification, users could carry their health data across borders and access healthcare services wherever they are in the world.

5.5. Blind Insurance Policies (BIPs) & AI Powered Risk Modeling

Salus is already working on Blind Insurance Policies (BIPs), which issue insurance purely based on health metrics and behaviors, without personal demographic data. The next step will be to use AI powered risk modeling to better predict health outcomes based on continuous health data inputs. By using machine learning algorithms, Salus can offer more accurate and personalized insurance policies that adapt in real time based on user behavior.

Future Vision: AI Driven Personalized Insurance

Imagine AI powered insurance products that not only adjust premiums based on health data but also provide personalized health recommendations to help users lower their premiums. For example, an AI model could predict that a user's health score will decrease based on current trends (e.g., less exercise or higher blood pressure) and suggest personalized interventions like changes in diet or exercise routine. Salus could also integrate these models with wearables and IoT devices to provide real time feedback and proactive healthcare recommendations, creating a fully integrated, predictive healthcare insurance loop.

5.6. Tokenized Wellness Ecosystem and Corporate Wellness Programs

Salus can expand into the corporate wellness space by offering tokenized incentives for employee health programs. Employers could integrate Salus into their benefits packages, where employees earn Salus tokens or insurance discounts by participating in health challenges, sharing fitness data, and engaging in corporate sponsored wellness initiatives.

Future Vision: Salus as a Corporate Wellness Platform

Salus could become the go-to platform for corporate wellness programs. Companies could sponsor challenges where employees earn tokens for improving their health, achieving fitness goals, or attending health related events. These tokens could be redeemed for corporate benefits, premium reductions, or wellness services like gym memberships. Tokenized wellness ecosystems could also include partnerships with fitness brands, mental health apps, and even nutrition services, allowing users to seamlessly earn and spend tokens in a fully integrated wellness environment.

5.7. Global Scaling and Cross Jurisdictional Health Systems

As blockchain technology and decentralized protocols gain wider acceptance, Salus can scale its model to operate across multiple jurisdictions and regulatory frameworks. This will allow users from different countries to leverage the platform, contributing their health data and benefiting from international insurance providers and researchers.

Future Vision: Salus as a Global Health Protocol

Salus could become a global health data sharing protocol that transcends national boundaries. With a focus on user control and privacy, the platform could integrate with cross border health systems, enabling users to share data globally while benefiting from local and international insurers. Salus could facilitate a global health score standard, which would be universally accepted by insurers and researchers alike, allowing for seamless cross border insurance coverage and research collaboration.

5.8. Expanded Use of Salus Tokens for Real World Services

As the Salus token economy matures, users could be able to spend tokens not only within the platform but also in the broader health and wellness industry. Imagine earning tokens for health data sharing and then spending those tokens on gym memberships, nutrition plans, or even mental health services.

Future Vision: Salus Token as a Global Health Currency

Salus tokens could evolve into a universal health currency, accepted by a wide range of health and wellness providers globally. Users could use their tokens to pay for doctor visits, telemedicine consultations, fitness classes, and even health related retail products. This would create a self-sustaining ecosystem where health data, wellness activities, and healthcare services are all interconnected, further incentivizing users to adopt healthy behaviors and actively engage with their health.

5.9. Long Term Vision: A Fully Decentralized Healthcare Ecosystem

In the long term, Salus could catalyze the development of a fully decentralized healthcare ecosystem. By integrating telemedicine, personalized diagnostics, and decentralized medical records, Salus could enable users to take full control of their healthcare journey – from prevention and diagnostics to treatment and insurance.

Future Vision: The Decentralized Healthcare Network (DHN)

Salus could be a foundational pillar of a Decentralized Healthcare Network (DHN), where users access healthcare services in a decentralized, borderless manner. Medical records could be stored on chains, allowing users to move seamlessly between healthcare providers across the globe, with insurers dynamically adjusting premiums based on real-time health data. Healthcare decisions would be user driven, with smart contracts ensuring that care is delivered fairly, transparently, and efficiently.

Certainly! Here’s how the section on the **Advanced Health Metric Synthesis Algorithm** can be integrated into the white paper:

6. Advanced Health Metric Synthesis Algorithm

As part of the Salus protocol's commitment to providing accurate and actionable health insights, we have developed an **Advanced Health Metric Synthesis Algorithm**. This algorithm synthesizes multiple health metrics into a comprehensive health score, taking into account the nuances of biological data and potential anomalies.

1. Feature Vector Normalization

The algorithm begins with a multi-dimensional feature vector \($F = [f_1, f_2, \ldots, f_7]$ \), where each \(f_i \) represents a distinct health metric (heart rate, steps, calories, active time, sleep time, blood oxygen, and blood pressure). To mitigate the impact of varying scales and units, we employ min-max normalization:

F_norm = [(f_i - min(F)) / (max(F) - min(F)) for f_i in F]

This transformation maps each feature to the interval [0, 1], preserving the relative distribution of the original data while ensuring commensurability across all dimensions.

2. Weighted Feature Aggregation via Neural Network Simulation

We implement a rudimentary simulation of a single-layer neural network by applying a weight vector \($W = [w_1, w_2, \ldots, w_7]$ \) to our normalized feature vector:

S = Σ(F_norm[i] * W[i]) for i in [1, 7]

This weighted sum \(S \) represents a linear combination of the input features, analogous to the pre-activation output of a neural network layer. The weights \(W \) are predetermined based on domain expertise but could potentially be optimized through gradient descent in a more sophisticated implementation.

3. Non-linear Activation

To introduce non-linearity into our model, emulating the complexity of biological systems, we apply the Rectified Linear Unit (ReLU) activation function:

A = max(0, S)

This operation introduces a simple non-linearity, allowing our model to capture more complex relationships between the input features and the output score.

4. Statistical Anomaly Detection

Concurrently, we perform a statistical analysis on the original feature vector to identify anomalous health metrics. We calculate the z-score for each feature:

$\mu = (1/n) * \Sigma f_i$ (arithmetic mean)

$\sigma = sqrt((1/n) * \Sigma(f_i - \mu)^2)$ (standard deviation)

$Z = [(f_i - \mu) / \sigma$ for f_i in $F]$

Z-scores quantify the number of standard deviations each feature is from the mean. We define an anomaly threshold \(\tau = 2 \), considering any \(|zᵢ| > \tau \) as anomalous.

5. Anomaly Penalization

We quantify the anomaly presence:

$\alpha = |\{z_i : |z_i| > \tau, z_i \in Z\}|$

This \(\alpha \) represents the count of features exceeding our anomaly threshold. We then define a penalty function:

$P(\alpha) = k * \alpha$

Where (k) is a hyperparameter (set to 0.1 in our implementation) controlling the severity of the penalty.

6. Final Score Computation

The final health score (H) is computed as:

$$H = \text{round}((A * 100 - P(\alpha)) * 10) / 10$$

This operation scales our activated sum, applies the anomaly penalty, and rounds to one decimal place, producing a final score that synthesizes multiple health metrics while accounting for statistical anomalies.

Algorithmic Complexity and Potential Enhancements

The current implementation has a time complexity of ($O(n)$), where (n) is the number of features, making it computationally efficient. However, several enhancements could significantly increase its sophistication:

1. Dynamic Weight Adjustment: Implement a backpropagation algorithm to optimize weights based on historical data and outcomes.

2. Multi-layer Perceptron: Extend the neural network simulation to include multiple layers with various activation functions (e.g., sigmoid, tanh) to capture more complex non-linear relationships.

3. Temporal Analysis: Incorporate time-series analysis to account for trends and cyclical patterns in health metrics.

4. Ensemble Methods: Combine multiple models (e.g., decision trees, support vector machines) with our current approach in an ensemble to improve robustness and accuracy.

5. Bayesian Inference: Implement a Bayesian framework to quantify uncertainty in our health score and provide confidence intervals.

These enhancements would transform our algorithm into a more comprehensive and statistically robust health metric synthesis system, capable of capturing intricate patterns and providing more nuanced health insights.

7. Conclusion

The Salus protocol is not just a solution for incentivized health data sharing; it represents a vision for the future of healthcare and insurance. Through its privacy-preserving architecture, tokenized incentives, and decentralized governance, Salus empowers individuals to control and monetize their health data while contributing to cutting-edge research and receiving insurance benefits. As Salus grows, it has the potential to reshape the way we think about health data, privacy, insurance, and healthcare services. With its commitment to transparency, security, and user empowerment, Salus is poised to lead the next generation of health innovation in the Web3 era.

By imagining a world where individuals have full control over their health data, are incentivized to lead healthier lives, and can access affordable and personalized insurance and healthcare services, Salus is laying the foundation for a healthier, more equitable future.